

3.4 Tales from the Haematology Clinic I

▼ What is anaemia?

- Anaemia is a **reduction in the amount of haemoglobin** in a given volume of blood below what would be expected in comparison with a healthy subject of the same age and gender
- **Reduction in haemoglobin concentration**

▼ What other visual characteristic is usually associated with microcytic erythrocytes?

- They are **hypochromic (pale)**

▼ Why is this the case?

- The microcytic cells usually have **less haemoglobin** and so are less red than normal erythrocytes
- The **central pallor of the microcytic cells is also typically much larger** (more than the typical 1/3 of area) and so this leads to a paler cell that has less haemoglobin

▼ What is a clinical sign seen in patients with low iron?

- **Fatigue**
- **Dizziness**

▼ What is ferritin?

- Ferritin is a protein that is used to **store iron**
 - It can be tested for when you want to do a test for iron levels

▼ What are the three common causes of iron deficiency anaemia?

- Increased blood loss
- Insufficient iron intake
- Increased iron requirements (i.e pregnancy)

▼ How can low iron be treated?

- Iron replacement therapy
 - i.e **Ferrous sulphate tablets**

▼ What is beta-thalassaemia?

- **Beta-thalassaemia** is an **autosomal recessive disorder** which **affects the blood**

▼ What happens in beta-thalassaemia?

- In **normal individuals**, the globin protein of haemoglobin consists of **two alpha and two beta chains**
 - This forms HbA
- In individuals with **beta-thalassaemia**, the globin protein of haemoglobin consists of **two alpha and two delta chains**
 - This forms HbA2

▼ Recall two common Full Blood Count characteristics seen in iron deficiency and beta-thalassaemia.

- Both will cause a **decrease in mean cell volume (MCV)**
- Both will cause a **decrease in mean cell haemoglobin** (albeit for different reasons)

▼ Recall three ways you can distinguish between iron deficiency and beta-thalassaemia.

- **Iron deficiency** will lead to **low ferritin**, beta-thalassaemia will not
- **Iron deficiency** will cause **lower total Haemoglobin (Hb)**, beta-thalassaemia will not
- **Beta-thalassaemia** will cause a **higher RBC count**, iron deficiency will cause a **lower RBC count**
- **Beta-thalassaemia** has **HbA2**, iron deficiency does not
 - Alpha-thalassaemia however, has normal HbA

▼ Image showing differences between iron deficiency and thalassaemia trait.

	<u>Iron deficiency</u>	<u>Thalassaemia trait</u>
Haemoglobin (Hb)	Normal or ↓ Low (in proportion to Hb) Low (in proportion to Hb) Low Low	Normal (or mildly ↓) Lower for same Hb Lower for same Hb Relatively preserved Increased
Electrophoresis	Normal	Hb A ₂ raised in β-thal trait*
Mean cell volume	Low	Normal

*Normal in α-thal

▼ What is transferrin?

- A **protein produced by the liver** that regulates iron absorption into the blood

▼ What causes changes in transferrin levels?

- High iron levels indicate **low transferrin levels**
- Low iron levels indicate **high transferrin levels**

▼ What could cause a defect in haem synthesis?

- Iron deficiencies

▼ What is the name given to the reduced synthesis of globin?

- Thalassaemia
- Beta-thalassaemia is when you have less beta chains of globin, and so haemoglobin synthesis is impaired
 - The alpha chains then combine with delta chains to compensate for this lack

▼ How is vitamin B12 absorbed in the body?

- Vitamin B12 combines with intrinsic factor made in the **gastric parietal cells** to form B12-IF
- B12-IF then binds to receptors in the ileum

▼ Recall three causes of vitamin B12 deficiency.

- Inadequate intake
- Malabsorption (i.e Coeliac disease)
- Inadequate secretion of intrinsic factor (pernicious anaemia)
- Lack of hydrochloric acid in the stomach (achlorhydria)

▼ What is a symptom associated with vitamin B12 deficiency?

- **Pins and needles**
- Fatigue (seen in all anaemia)
- **Vitamin B12 and folate** is required for the essential **functioning of the central nervous system (CNS)**

▼ How do sickle cells form?

- They result from the polymerisation of Haemoglobin S, which is less soluble than haemoglobin A
 - Only happens in deoxyhaemoglobin

▼ How is haemoglobin S formed?

- Haemoglobin S is formed when one or more copies of an abnormal beta-globin chain are inherited

▼ What is polychromasia?

- It is when there is an **increased blue 'tinge'** to an erythrocyte (polychromasia - many colours)
- It indicates that the red cell is young and reflects increased erythropoiesis as a result of haemolysis

▼ What is another special consideration about polychromatic cells?

- The cells are typically larger than normal red blood cells.

▼ What are target cells?

- **Erythrocytes** with an **accumulation of haemoglobin in the central pallor**.
 - i.e they have a "target" in the middle due to this haemoglobin



▼ Describe the mutation that occurs in erythrocytes that leads to sickle cell anaemia.

- A **negatively charged glutamate** residue is replaced by an **uncharged valine residue** at **position 6** on the **beta-globin chain**
- This makes the (haemo)globin more **hydrophobic**, and hence **sticky**.

▼ Where does this mutation occur?

- On the beta-globin chain

▼ What two medications do patients with sickle cell disease need to take?

- **Penicillin V**
- **Folic acid**

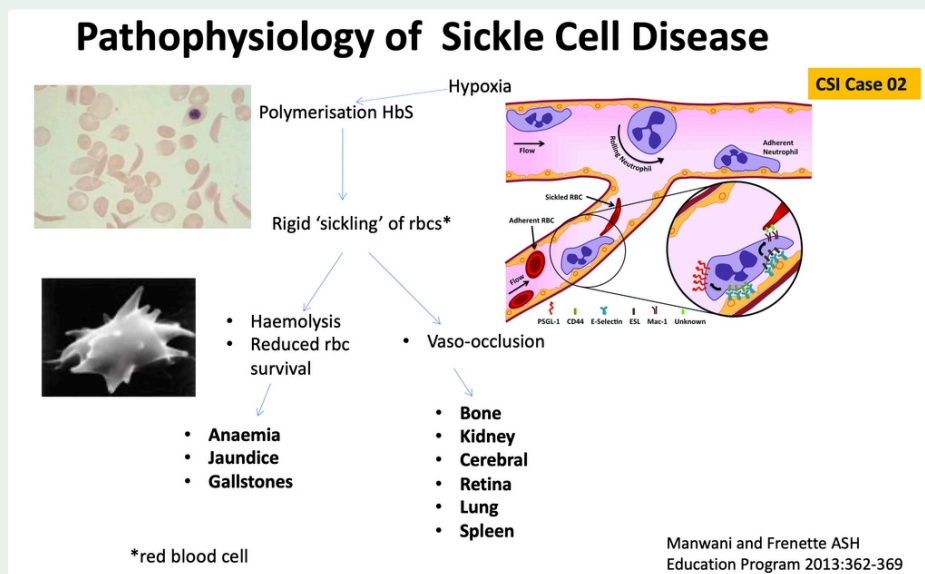
- ▼ Why do people with sickle cell anaemia need to take Penicillin V?
 - Patients with HbSS are **functionally hyposplenic** due to infarction of the spleen, usually by the age of 5 years. This puts them at **increased risk of infections by encapsulated bacteria**.
 - The spleen is involved in immune defence
 - To reduce this risk they **require vaccination** against and to take **daily prophylactic (preventative) antibiotics**
- ▼ Why do people with sickle cell anaemia need to take folic acid?
 - In **haemolytic anaemias such as HbSS** the **increased red blood cell turnover** and production **increases the folic acid requirement**.
 - Folic acid is required for erythropoiesis
- ▼ What is a megaloblast?
 - An abnormal erythroblast that is larger than normal.
- ▼ What is megaloblastic anaemia?
 - Anaemia that is caused by vitamin B12/folic acid deficiencies
- ▼ What is haemolytic anaemia?
 - A condition where **red blood cells are destroyed at a much faster rate** than they are replaced.
- ▼ What is a reticulocyte?
 - An immature blood cell that can develop into a mature one (erythrocyte)
 - Levels can increase with red cell production, as more erythrocytes are being developed.
- ▼ What is reticulocytosis and how is it caused?
 - It is an increased number of reticulocytes
 - It may be due to increased bleeding or erythrocyte destruction (haemolysis)
- ▼ What is a poikilocyte?
 - A red blood cell that has an abnormal shape.
- ▼ What is the distinct property of Haemoglobin S that impairs its function?
 - It has a much **lower affinity for oxygen than haemoglobin A**

▼ What are Howell Jolly Bodies?

- **Remnants of DNA** from nuclei that **haven't been fully expelled by erythroblasts during their maturation** in the bone marrow

▼ Describe the pathophysiology of sickle cell disease.

- **Deoxyhaemoglobin** is present **when O₂ has been released from haemoglobin to tissues**
 - Gets worse in states of hypoxia, such as exercise
- This causes **polymerisation of HbSS** within red blood cells, leading to "sickle cells" forming
- This causes **increased intravascular haemolysis** and **vaso-occlusion**
- **Severe intravascular haemolysis** can lead to anaemia, gallstones, jaundice and folic acid deficiency
- **Vaso-occlusion** can lead to infarctions of various tissue such as brain, spleen, kidney, bone, retina and lung



▼ What effect does hookworm have on sufferers?

- It is an infection that feeds on the blood, hence reducing blood levels.

▼ What is haemoglobin A2?

- It is an abnormal form of adult haemoglobin that typically forms in larger amounts in thalassaemia sufferers.

▼ What is hepcidin?

- A protein that is used to regulate iron levels in the body.

▼ When are hepcidin levels increased/reduced?

- **Increases in iron levels** in the plasma and iron storage **stimulate the production and release of hepcidin from hepatocytes**
- This **blocks further iron absorption from the diet** and its **further storage** in the small intestine
- When there is a need for **erythropoiesis**, **hepcidin levels are reduced** and iron absorption is increased into these cells.

▼ Describe the mechanism through which hepcidin works to inhibit iron absorption.

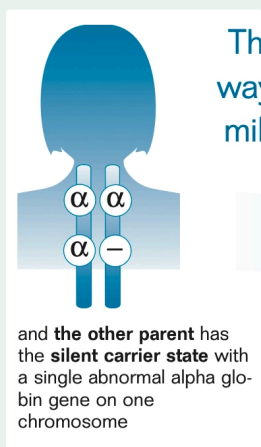
- **Hepcidin binds to its receptor ferroportin**, which **prevents the efflux of iron from duodenal enterocytes (intestinal cells) into the blood** after absorption from the small intestine
 - The **enterocytes eventually die** and the **iron is lost into the gut lumen and not absorbed into the blood**

▼ How many genes do we have that code for alpha globin?

- **Four**

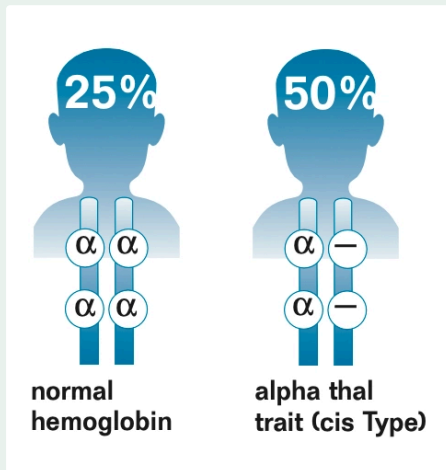
▼ In alpha thalassaemia, what is the silent carrier state?

- An individual with **one abnormal alpha globin gene** is said to be a **silent carrier** of alpha thalassemia
- They will typically not have any health problems



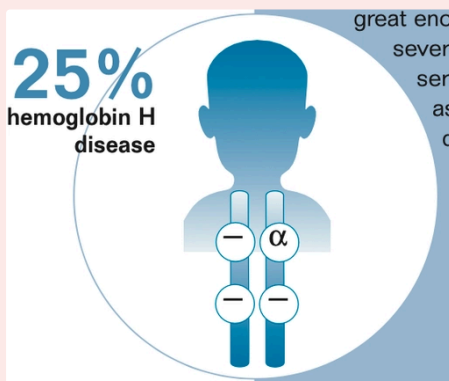
▼ In alpha thalassaemia, what is the thalassaemia trait?

- An individual with **two abnormal alpha globin genes** is said to have **thalassaemia trait**
- They can either have the cis trait, on the same chromosome, or the trans trait, on different chromosomes

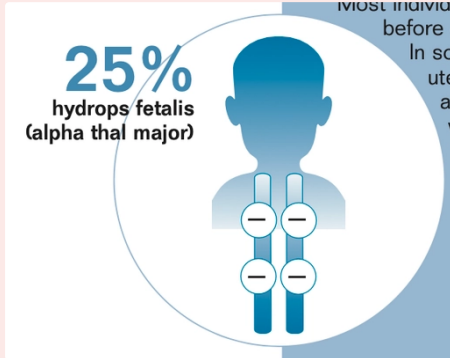


▼ In alpha thalassaemia, what is Haemoglobin H disease?

- An individual with **three abnormal alpha globin genes** is said to have **Haemoglobin H disease**
- This can result in **splenomegaly, bone deformities and fatigue**



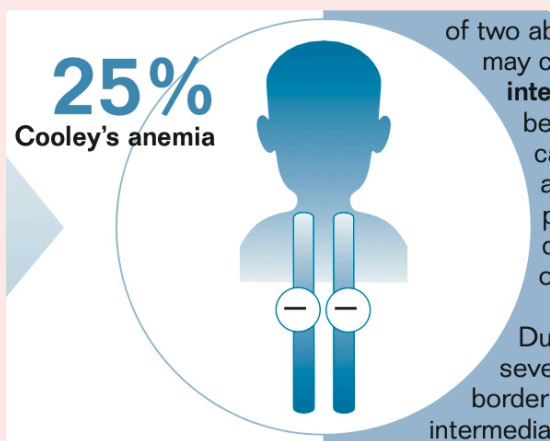
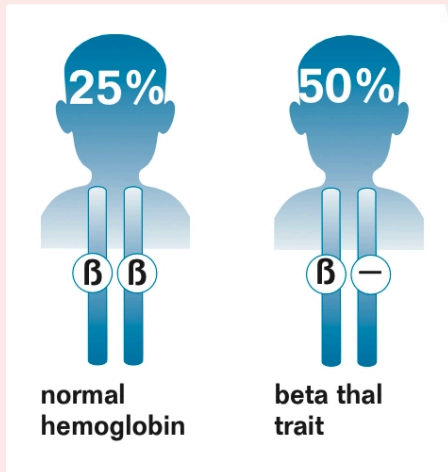
- ▼ In alpha thalassaemia, what is the hydrops fetalis/alpha thalassaemia major?
- An individual with **all four abnormal alpha globin genes** is said to have **hydrops fetalis**
 - Most individuals with this condition **die shortly after birth**



- ▼ How many genes do we have that code for beta globin?
- **Two.**

▼ What are the two types of beta thalassaemia?

- **Beta thalassaemia trait/minor** where no real health problems are experienced
- **Beta thalassaemia major (Cooley's anaemia)** where the lack of beta-globin causes a life-threatening anaemia and **haemoglobin function is greatly impaired**



▼ What is haemoglobin E trait?

- This is a **mutant form of beta-globin**

▼ What is E beta thalassaemia?

- When **combined with the beta-thalassaemia trait**
- This leads to **E beta thalassaemia**

